

# Sign Local System Through the Finite-Adele Boundary

Autonomous discussion Pass 102 pushes the Pass-101 sign local system through the primitive collapse and the all-prime constant-term boundary.

After a primitive collapse  $T_S \rightarrow \mathbb{Q}/\mathbb{Z}$ , the one-generator unit extension is

$$\beta = [0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0].$$

The all-prime constant-term row is

$$\delta_\epsilon = [0 \rightarrow \mathbb{Q} \rightarrow \mathbb{A}_f \rightarrow \epsilon \rightarrow 0], \quad \epsilon = \mathbb{A}_f/\mathbb{Q}.$$

The  $\mathbb{Z}/2$  sign local system acts by multiplying these boundary/Yoneda classes:

$$\beta \mapsto \sigma\beta, \quad \delta_\epsilon \mapsto \sigma\delta_\epsilon, \quad \sigma \in \{\pm 1\}.$$

Thus the sign is not a new boundary object. It is the coefficient system on the one-dimensional boundary line spanned by the extension class.

At finite level  $N$ , the unit extension has Bockstein shadow

$$1 \in \text{Ext}^1(\mathbb{Z}/N, \mathbb{Z}) \cong \mathbb{Z}/N.$$

The signed class is  $\pm 1 \in \mathbb{Z}/N$ . This sign is visible exactly when  $N > 2$  and collapses for  $N = 2$ .

The finite-adele boundary morphism

$$\partial_\epsilon : \epsilon \rightarrow \mathbb{Q}[1]$$

is the shifted class behind

$$D\epsilon \simeq \mathbb{Q}[-1].$$

A one-sided sign change negates this class. Applying the sign on both source and target multiplies the class by  $(-1)^2 = 1$ , so biduality remains involutive.

The coarse antipode quotient alone is still insufficient because it forgets whether the class is  $+\delta_\epsilon$  or  $-\delta_\epsilon$ . The  $\mathbb{Z}/2$  local system is sufficient for the finite sign bookkeeping, and it does not create a degree-0 Weyl/Fourier map  $\epsilon \rightarrow \mathbb{Q}$ .

The next task is to package this signed boundary class as a natural transformation over finite conductor reductions and compare it with the CRT-acyclic finite constant-term complexes.

The checker `artifacts/reports/pass102-sign-local-system-adele-boundary-check.json` verifies finite Bockstein sign classes, signed primitive collapse surjectivity, support-transport preservation, one-sided Yoneda negation, two-sided biduality involutivity, and sufficiency of the sign local system.